STEP 1 : Open Google Colab

STEP 2 — Install Required Libraries

```
# Install required libraries (Run only if not already installed)

!pip install spacy
!pip install transformers
!pip install torch
!pip install pandas
!python -m spacy download en_core_web_sm
```

```
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-slim
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer>=0.2
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Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.4
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Requirement already satisfied: typer>=0.24.0 in /usr/local/lib/python3.12/dist-packages (from typer-slim->transformers) (0.24.0
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Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,
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Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->hugg
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-slim->transfo
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Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer>=0
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3.0
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.10.0+cpu)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch) (3.24.2)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (1.3.0
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch) (3.0.3)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.0.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.3)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17
Collecting en-core-web-sm==3.8.0
  Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0-py3-none-a
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 12.8/12.8 MB 89.3 MB/s eta 0:00:00
✓ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in
order to load all the package's dependencies. You can do this by selecting the
'Restart kernel' or 'Restart runtime' option.
```

STEP 3 — Import Libraries

```
# spaCy library for Natural Language Processing
import spacy

# displacy is used to visualize named entities
```

```
from spacy import displacy

# Hugging Face tools for loading transformer models
from transformers import pipeline, AutoTokenizer, AutoModelForTokenClassification

# PyTorch backend for transformer models
import torch

# pandas for storing results in table format
import pandas as pd
```

STEP 4 — Download and Load spaCy Model

```
# Load small English pretrained model
nlp = spacy.load("en_core_web_sm")
```

STEP 5 — Prepare Sample Text Data

```
# Create real-world example sentences

sentences = [
    "Apple launched the new iPhone 15 in California.",
    "Lionel Messi scored a goal for Inter Miami in the MLS final.",
    "The Prime Minister of India Narendra Modi visited France.",
    "Microsoft announced a partnership with OpenAI in San Francisco.",
    "NASA plans to send astronauts to Mars by 2030."
]
```

STEP 6 — Apply spaCy NER

```
print("========== spaCy NER Results ==========\n")

for sentence in sentences:
    doc = nlp(sentence)  # Process sentence through spaCy pipeline

    print(f"Sentence: {sentence}")

    for ent in doc.ents:
        # ent.text = entity name
        # ent.label_ = entity category
        print(f"Entity: {ent.text} | Label: {ent.label_}")

    print("-" * 50)
```

```
========== spaCy NER Results ==========

Sentence: Apple launched the new iPhone 15 in California.
Entity: Apple | Label: ORG
Entity: 15 | Label: CARDINAL
Entity: California | Label: GPE
--------------------------------------------------
Sentence: Lionel Messi scored a goal for Inter Miami in the MLS final.
Entity: Lionel Messi | Label: PERSON
Entity: MLS | Label: ORG
--------------------------------------------------
Sentence: The Prime Minister of India Narendra Modi visited France.
Entity: France | Label: GPE
--------------------------------------------------
Sentence: Microsoft announced a partnership with OpenAI in San Francisco.
Entity: Microsoft | Label: ORG
Entity: OpenAI | Label: GPE
Entity: San Francisco | Label: GPE
--------------------------------------------------
Sentence: NASA plans to send astronauts to Mars by 2030.
Entity: NASA | Label: ORG
Entity: Mars | Label: LOC
Entity: 2030 | Label: DATE
--------------------------------------------------
```

STEP 7 — Store spaCy Results in Table

```
# Create empty list to store data
spacy_data = []
```

```
for sentence in sentences:
    doc = nlp(sentence)

    for ent in doc.ents:
        spacy_data.append([sentence, ent.text, ent.label_])

# Create DataFrame
spacy_df = pd.DataFrame(spacy_data, columns=["Sentence", "Entity", "Label"])

# Display table
spacy_df
```

	Sentence	Entity	Label
0	Apple launched the new iPhone 15 in California.	Apple	ORG
1	Apple launched the new iPhone 15 in California.	15	CARDINAL
2	Apple launched the new iPhone 15 in California.	California	GPE
3	Lionel Messi scored a goal for Inter Miami in ...	Lionel Messi	PERSON
4	Lionel Messi scored a goal for Inter Miami in ...	MLS	ORG
5	The Prime Minister of India Narendra Modi visi...	France	GPE
6	Microsoft announced a partnership with OpenAI ...	Microsoft	ORG
7	Microsoft announced a partnership with OpenAI ...	OpenAI	GPE
8	Microsoft announced a partnership with OpenAI ...	San Francisco	GPE
9	NASA plans to send astronauts to Mars by 2030.	NASA	ORG
10	NASA plans to send astronauts to Mars by 2030.	Mars	LOC
11	NASA plans to send astronauts to Mars by 2030.	2030	DATE

Next steps: Generate code with spacy_df New interactive sheet

STEP 8 — (Optional) Visualize Entities

```
# Visualize NER for first sentence
doc = nlp(sentences[0])

displacy.render(doc, style="ent", jupyter=True)
```

Apple ORG launched the new iPhone 15 CARDINAL in California GPE .

STEP 9 — Load Hugging Face NER Model

```
# Load tokenizer
tokenizer = AutoTokenizer.from_pretrained("dbmdz/bert-large-cased-finetuned-conll03-english")

# Load model
model = AutoModelForTokenClassification.from_pretrained("dbmdz/bert-large-cased-finetuned-conll03-english")

# Create NER pipeline
ner_pipeline = pipeline("ner", model=model, tokenizer=tokenizer)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
config.json: 100%                                                    998/998 [00:00<00:00, 14.5kB/s]

tokenizer_config.json: 100%                                          60.0/60.0 [00:00<00:00, 984B/s]

vocab.txt:        213k/? [00:00<00:00, 3.10MB/s]

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster d
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to en
model.safetensors: 100%                                              1.33G/1.33G [00:15<00:00, 82.1MB/s]

Loading weights: 100%                                                391/391 [00:00<00:00, 607.49it/s, Materializing param=classifier.weight]

BertForTokenClassification LOAD REPORT from: dbmdz/bert-large-cased-finetuned-conll03-english
Key                       | Status     |  |
-------------------------+------------+--+-
bert.pooler.dense.weight | UNEXPECTED |  |
bert.pooler.dense.bias   | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
```

STEP 10 — Apply Transformer NER

```
print("========== Hugging Face NER Results ==========\n")

for sentence in sentences:
    results = ner_pipeline(sentence)

    print(f"Sentence: {sentence}")

    for res in results:
        word = res['word']
        label = res['entity']
        score = res['score']

        print(f"Word: {word} | Label: {label} | Confidence: {round(score,4)}")

    print("-" * 50)
```

```
========== Hugging Face NER Results ==========

Sentence: Apple launched the new iPhone 15 in California.
Word: Apple | Label: I-ORG | Confidence: 0.9965000152587891
Word: iPhone | Label: I-MISC | Confidence: 0.9958000183105469
Word: 15 | Label: I-MISC | Confidence: 0.994700014591217
Word: California | Label: I-LOC | Confidence: 0.9997000098228455
--------------------------------------------------
Sentence: Lionel Messi scored a goal for Inter Miami in the MLS final.
Word: Lionel | Label: I-PER | Confidence: 0.9991000294685364
Word: Me | Label: I-PER | Confidence: 0.9991999864578247
Word: ##ssi | Label: I-PER | Confidence: 0.998199999332428
Word: Inter | Label: I-ORG | Confidence: 0.9988999962806702
Word: Miami | Label: I-ORG | Confidence: 0.9645000100135803
Word: MLS | Label: I-ORG | Confidence: 0.921500027179718
--------------------------------------------------
Sentence: The Prime Minister of India Narendra Modi visited France.
Word: India | Label: I-LOC | Confidence: 0.9983000159263611
Word: Na | Label: I-PER | Confidence: 0.9993000030517578
Word: ##ren | Label: I-PER | Confidence: 0.9959999918937683
Word: ##dra | Label: I-PER | Confidence: 0.9988999962806702
Word: Mo | Label: I-PER | Confidence: 0.9991000294685364
Word: ##di | Label: I-PER | Confidence: 0.9861999750137329
Word: France | Label: I-LOC | Confidence: 0.9998000264167786
--------------------------------------------------
Sentence: Microsoft announced a partnership with OpenAI in San Francisco.
Word: Microsoft | Label: I-ORG | Confidence: 0.9995999932289124
Word: Open | Label: I-ORG | Confidence: 0.9986000061035156
Word: ##A | Label: I-ORG | Confidence: 0.9937999844551086
Word: ##I | Label: I-ORG | Confidence: 0.9975000023841858
Word: San | Label: I-LOC | Confidence: 0.9994999766349792
Word: Francisco | Label: I-LOC | Confidence: 0.9993000030517578
--------------------------------------------------
Sentence: NASA plans to send astronauts to Mars by 2030.
Word: NASA | Label: I-ORG | Confidence: 0.9975000023841858
```

```
Word: Mars | Label: I-LOC | Confidence: 0.9973999857902527
--------------------------------------------------
```

STEP 11 — Clean Transformer Output (Merge Subwords)

```
hf_data = []

for sentence in sentences:
    results = ner_pipeline(sentence)

    merged_entities = []
    current_entity = ""
    current_label = ""
    current_score = 0

    for res in results:
        word = res['word']
        label = res['entity']
        score = res['score']

        # Remove subword token indicator ##
        word = word.replace("##", "")

        # If continuation of same entity
        if label.startswith("I-") and current_entity:
            current_entity += " " + word
        else:
            if current_entity:
                hf_data.append([sentence, current_entity, current_label, current_score])

            current_entity = word
            current_label = label
            current_score = round(score,4)

    # Append last entity
    if current_entity:
        hf_data.append([sentence, current_entity, current_label, current_score])

# Create DataFrame
hf_df = pd.DataFrame(hf_data, columns=["Sentence", "Entity", "Label", "Confidence Score"])

hf_df
```

	Sentence	Entity	Label	Confidence Score
0	Apple launched the new iPhone 15 in California.	Apple iPhone 15 California	I-ORG	0.9965
1	Lionel Messi scored a goal for Inter Miami in ...	Lionel Me ssi Inter Miami MLS	I-PER	0.9991
2	The Prime Minister of India Narendra Modi visi...	India Na ren dra Mo di France	I-LOC	0.9983
3	Microsoft announced a partnership with OpenAI ...	Microsoft Open A I San Francisco	I-ORG	0.9996
4	NASA plans to send astronauts to Mars by 2030.	NASA Mars	I-ORG	0.9975

Next steps: Generate code with hf_df New interactive sheet

STEP 12 — Comparison Table

```
comparison = pd.DataFrame({
    "Feature": ["Model Type", "Speed", "Accuracy (qualitative)", "Context Handling", "Confidence Score"],
    "spaCy": [
        "Statistical + rule-based",
        "Very Fast",
        "Good",
        "Moderate",
        "No"
    ],
    "Hugging Face": [
        "Transformer (BERT)",
        "Slower",
        "Very High",
        "Excellent",
        "Yes"
    ]
```

```
})

comparison
```

	Feature	spaCy	Hugging Face
0	Model Type	Statistical + rule-based	Transformer (BERT)
1	Speed	Very Fast	Slower
2	Accuracy (qualitative)	Good	Very High
3	Context Handling	Moderate	Excellent
4	Confidence Score	No	Yes

Next steps: Generate code with comparison New interactive sheet